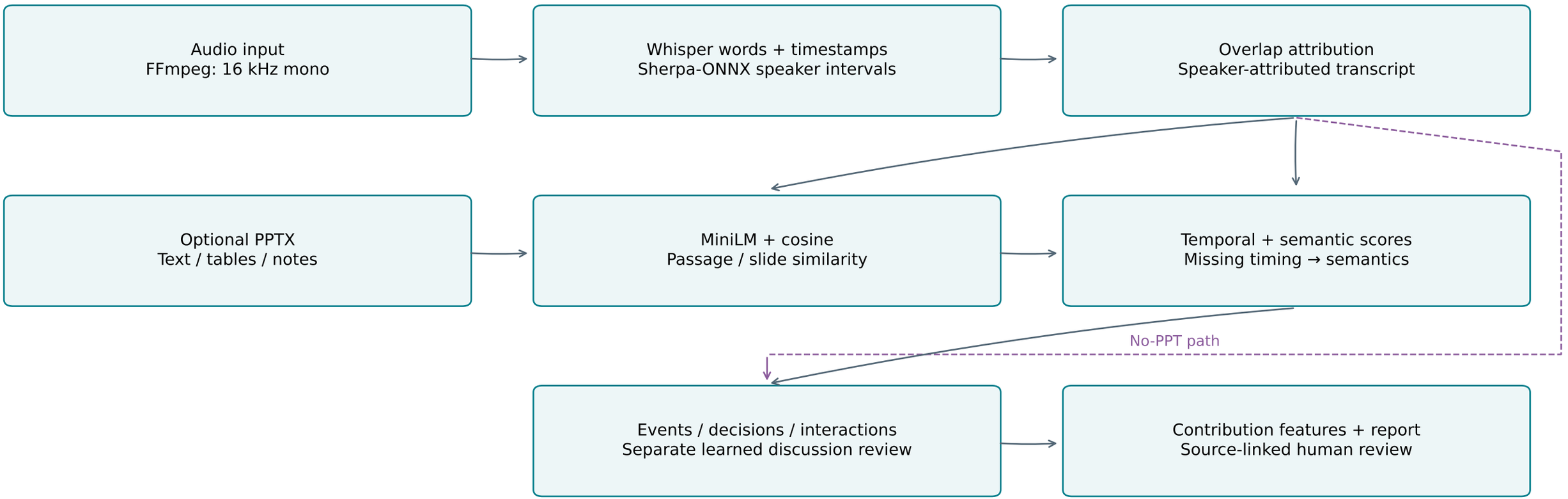
Implemented models and information flow



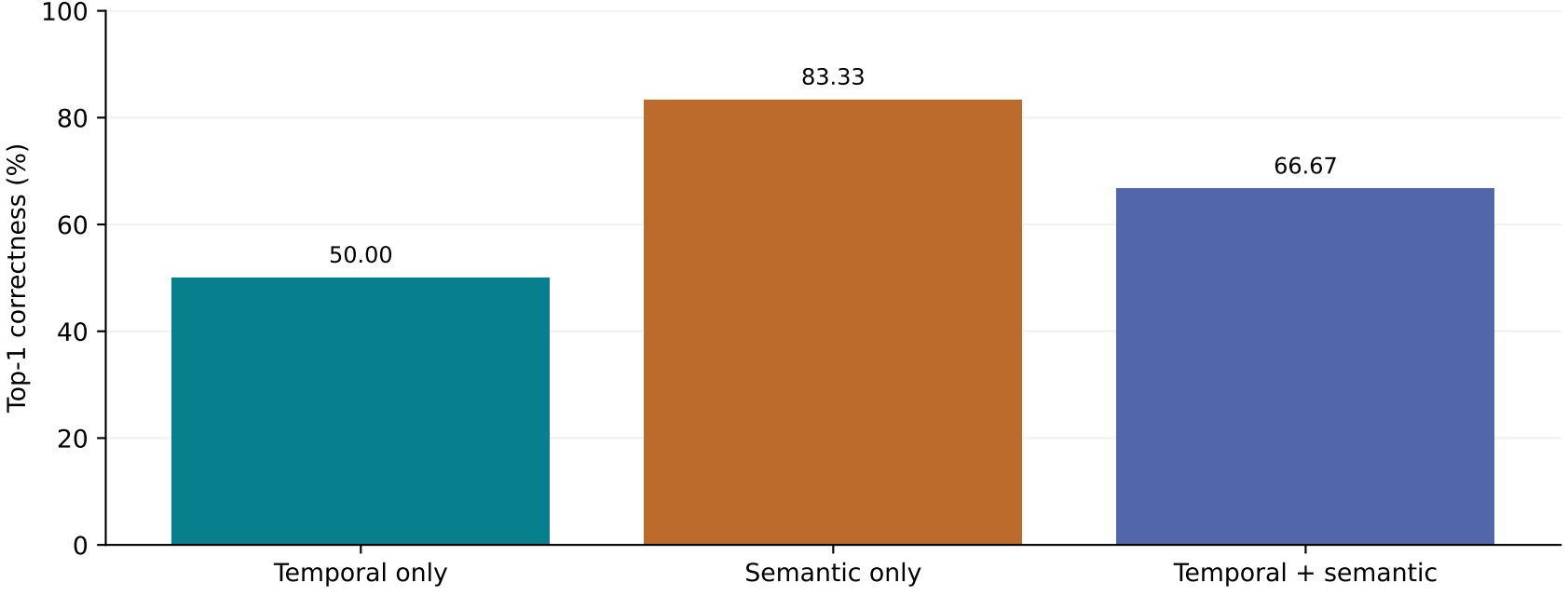
Conceptual dependency diagram, not measured performance. Speech and document contents are related through derived representations; no jointly trained raw-audio/image fusion network.

Which fusion methods have actually been used?

Information / method	What operation is this?	Current implementation
Word + speaker intervals	Temporal overlap association	Implemented; audio-derived information
Temporal + semantic evidence	Weighted score-level fusion	Implemented; optional timing; 0.5 / 0.5
Word + character + context TF-IDF	Feature concatenation	Implemented inside text classifier
Forest + logistic probabilities	Classifier-output ensemble	Implemented inside text; 0.75 / 0.25
MiniLM versus TF-IDF	Alternative representations	Selectable; not fused together
Participation + contribution features	Research feature combination	Implemented in GAP ranking; no slide input
Early / intermediate audio–document network	Joint learned multimodal fusion	Not deployed; separate toy prototypes exist
Cross-attention / tensor / learned graph fusion	Learned interaction mechanisms	Not implemented
Learned reliability gating	Input-dependent weighting	Not implemented; fixed missing-cue rule only

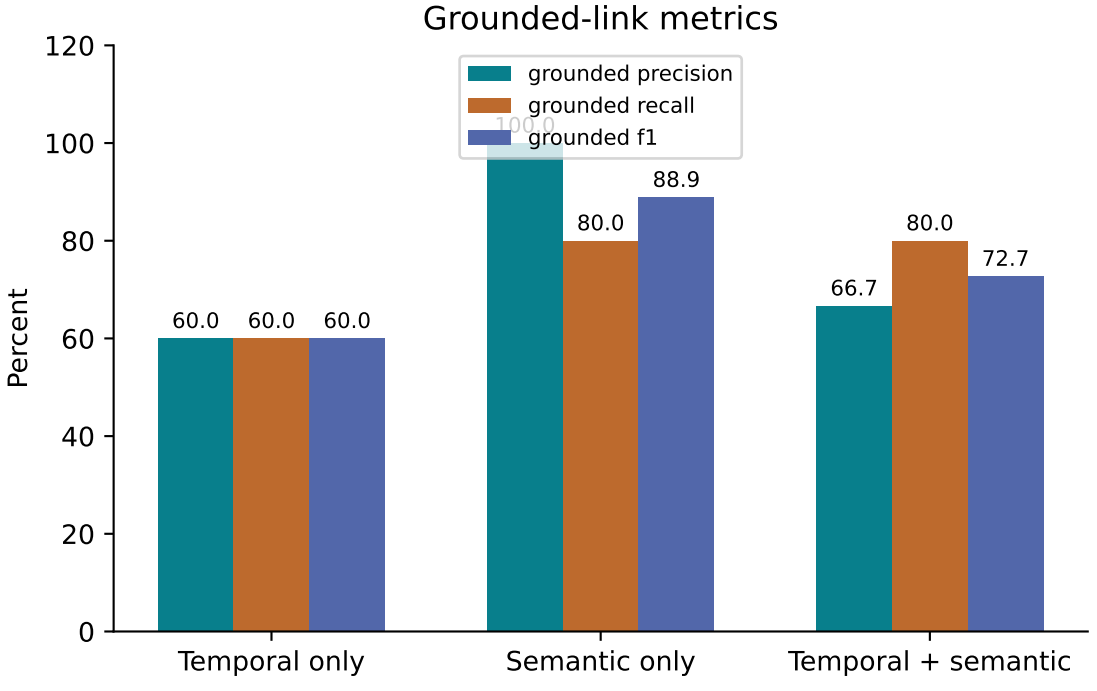
The temporal–semantic hybrid combines cues at the score level. It is not an architecture combining early and late neural fusion. Within-text concatenation/ensembling does not create extra input modalities.

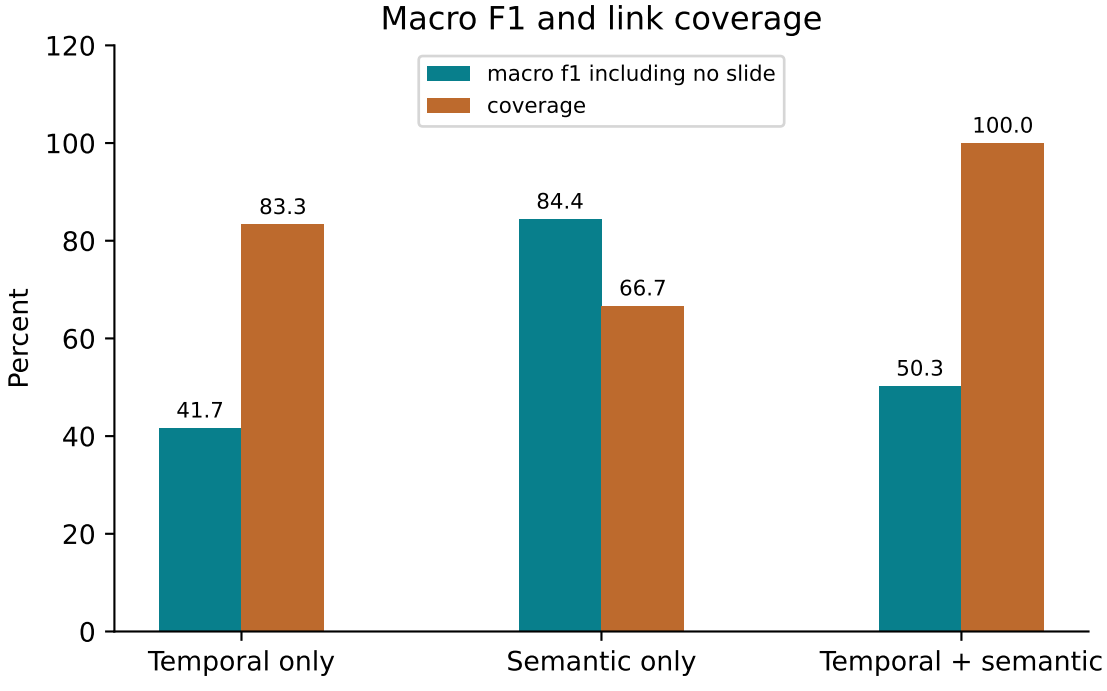
Controlled fusion diagnostic: semantic-only performs best



24 author-designed cases / six constructed slides. Expected NO_CONFIDENT_SLIDE is included in correctness. This is not independent real-meeting accuracy; no claim of universal hybrid superiority.

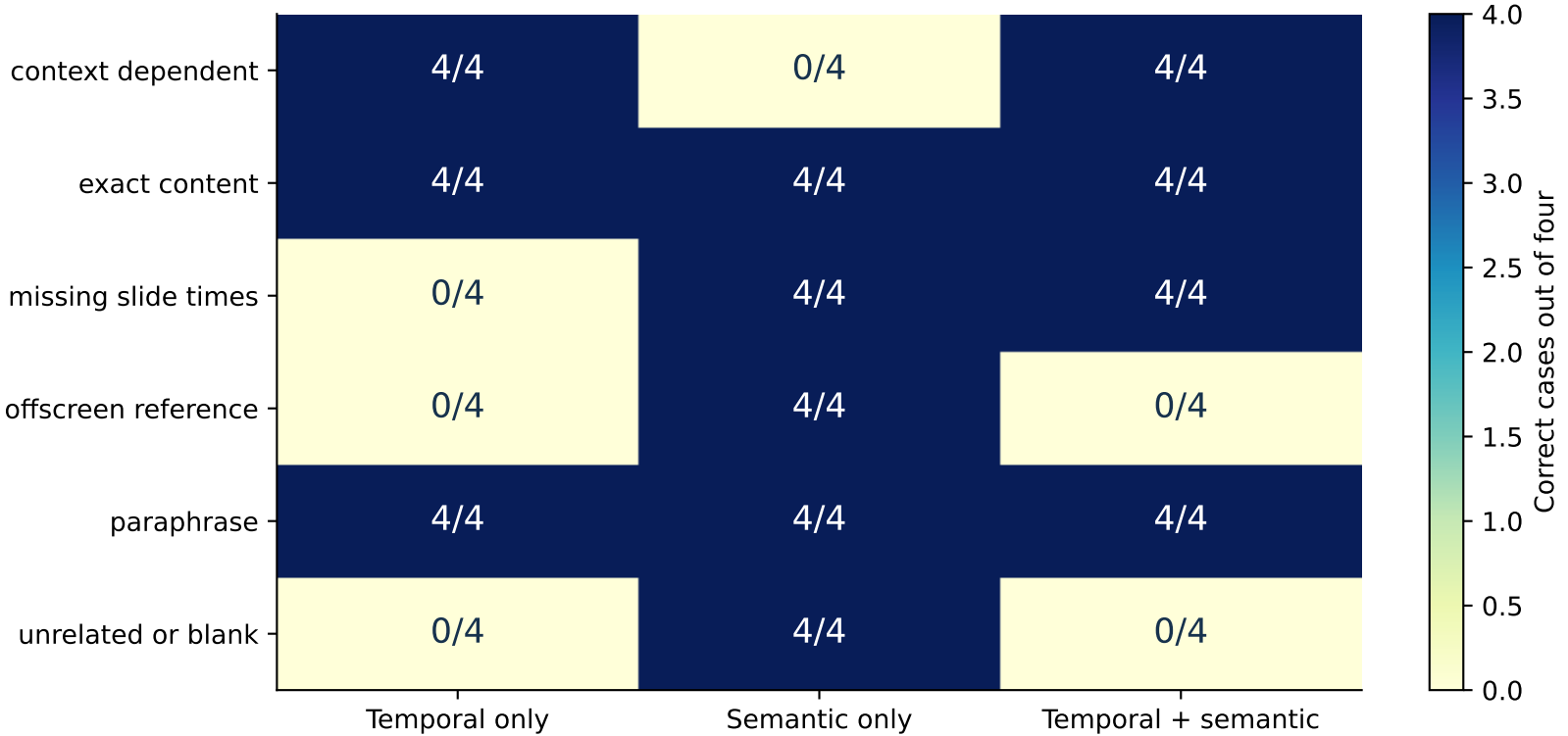
Fusion quality and coverage measure different things





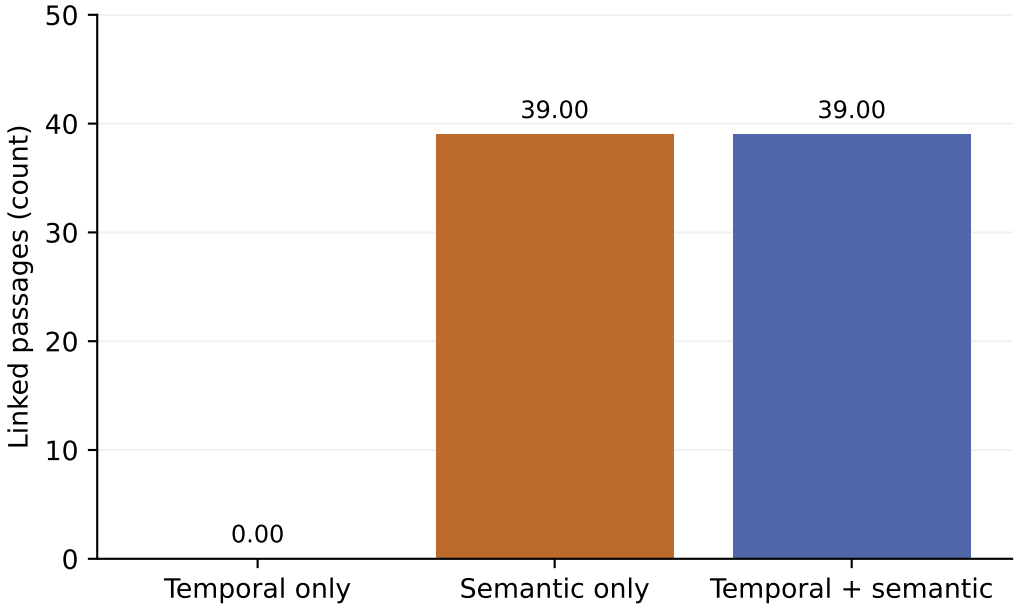
Constructed diagnostic only. Macro F1 includes the no-slide class. Hybrid coverage is 100% because it also links unrelated cases; higher coverage is not automatically better.

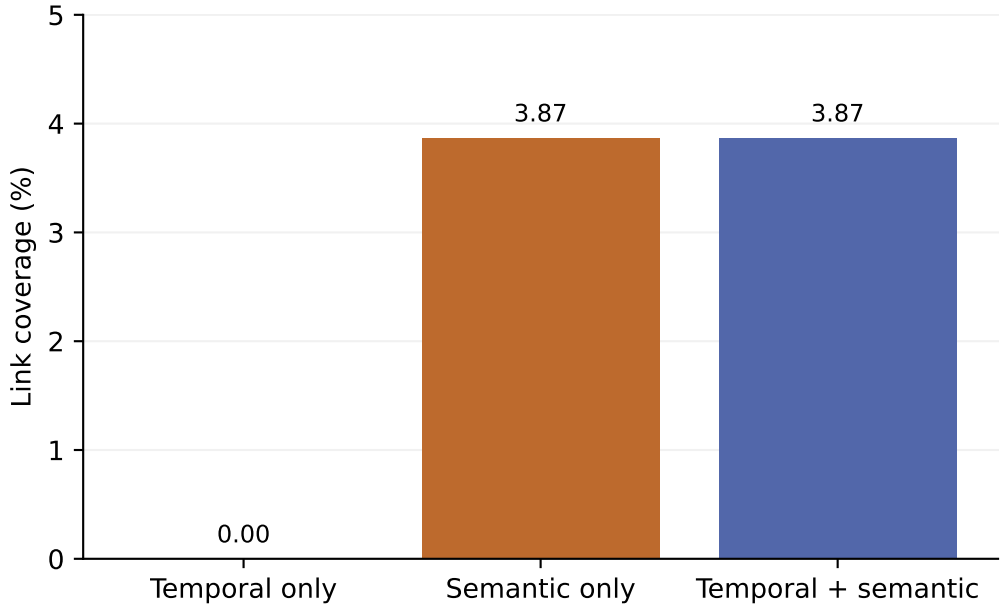
Where each alignment strategy succeeds and fails



Four cases per constructed group. Timing helps vague references but hurts offscreen/unrelated cases under the fixed 0.5/0.5 score rule and threshold 0.3.

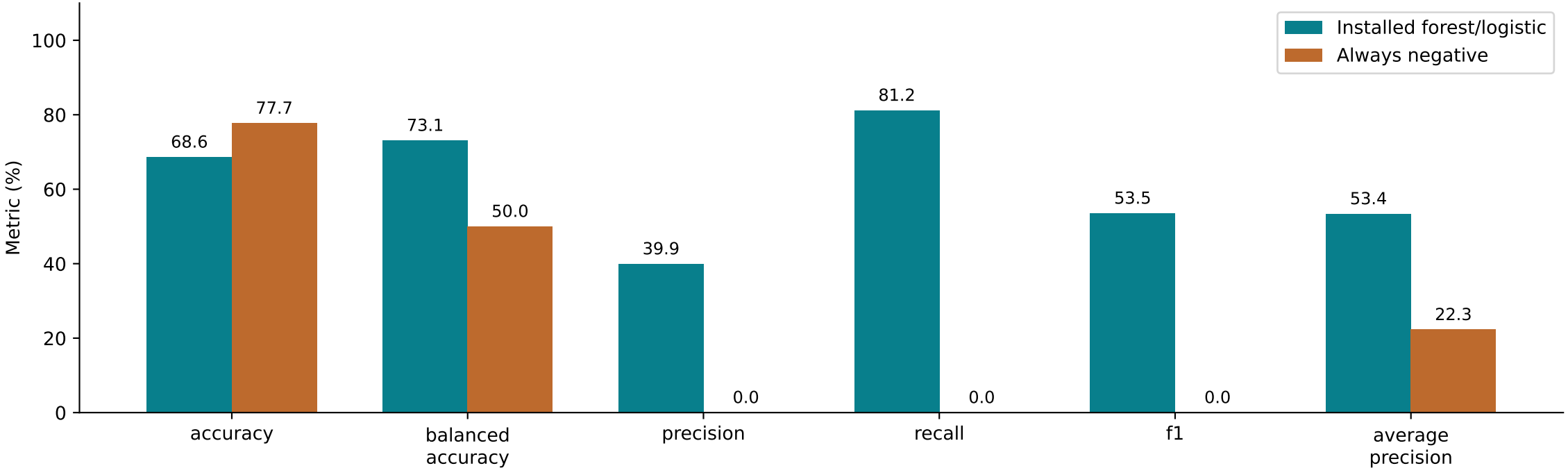
Real example: 39 / 1,008 passages linked; accuracy unknown





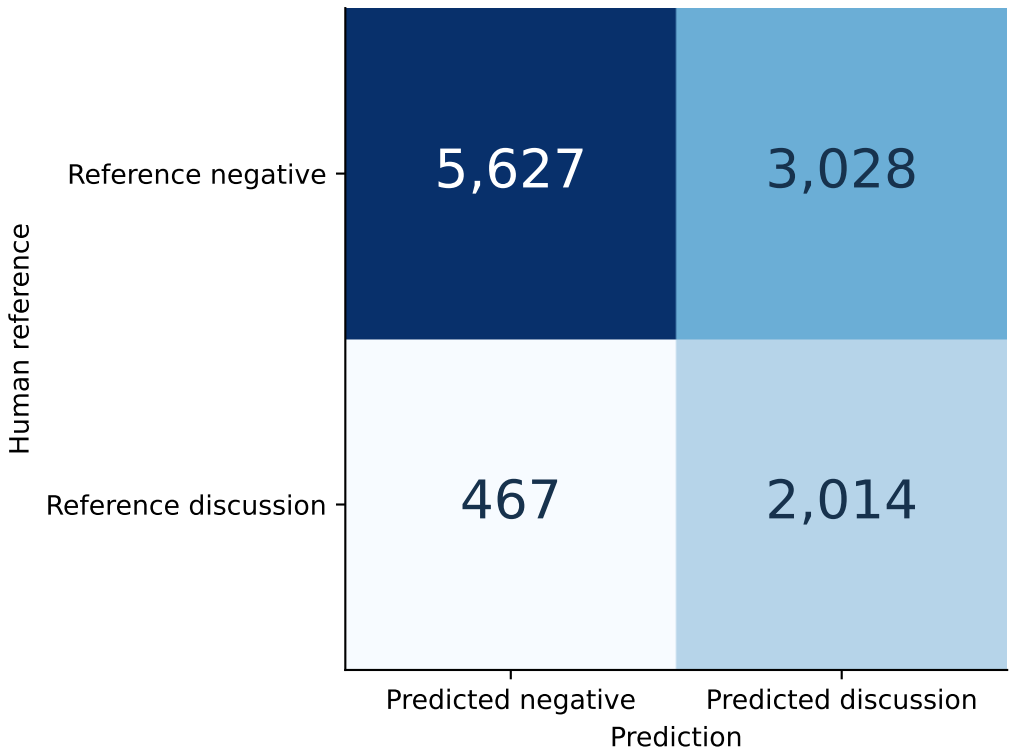
1008 passages / 34 slides / 0 valid timed slides. Semantic and hybrid agree because of the missing-timing fallback. Independent relevance labels are unavailable; coverage is not accuracy.

Discussion review: high recall, limited precision



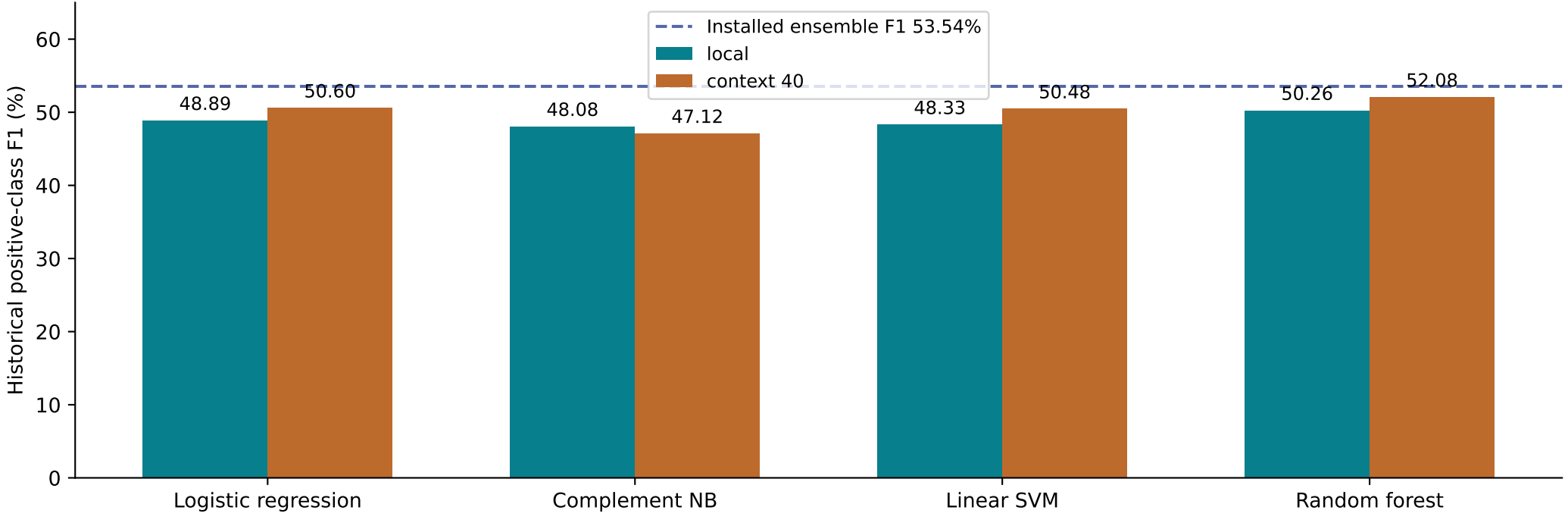
11,136 oracle AMI acts, 2,481 positives, 12 meetings / 3 project series; reused historical cohort. Always-negative accuracy is high because of imbalance, but it retrieves no positives. These results do not measure final decisions or raw-audio application accuracy.

Installed discussion classifier: historical confusion counts

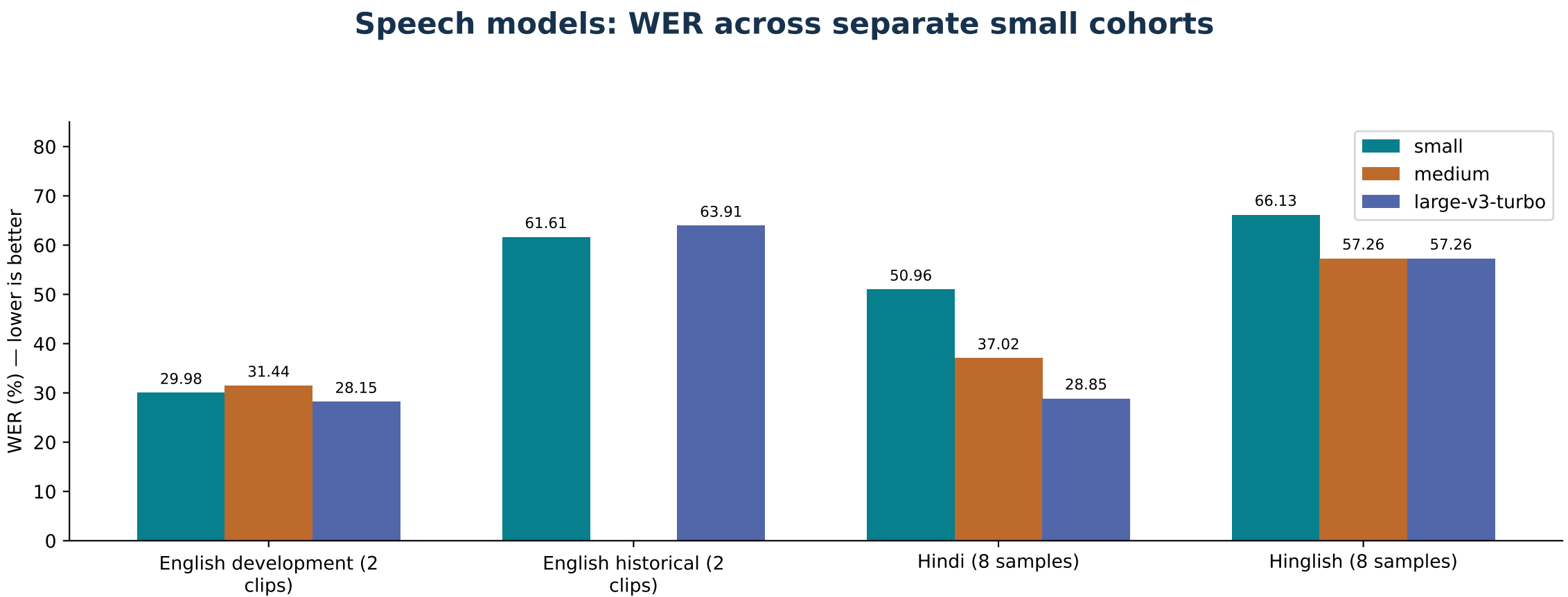


11,136 human reference dialogue acts: TN 5,627; FP 3,028; FN 467; TP 2,014. Reference text/boundaries differ from deploy
Historical cohort was repeatedly inspected.

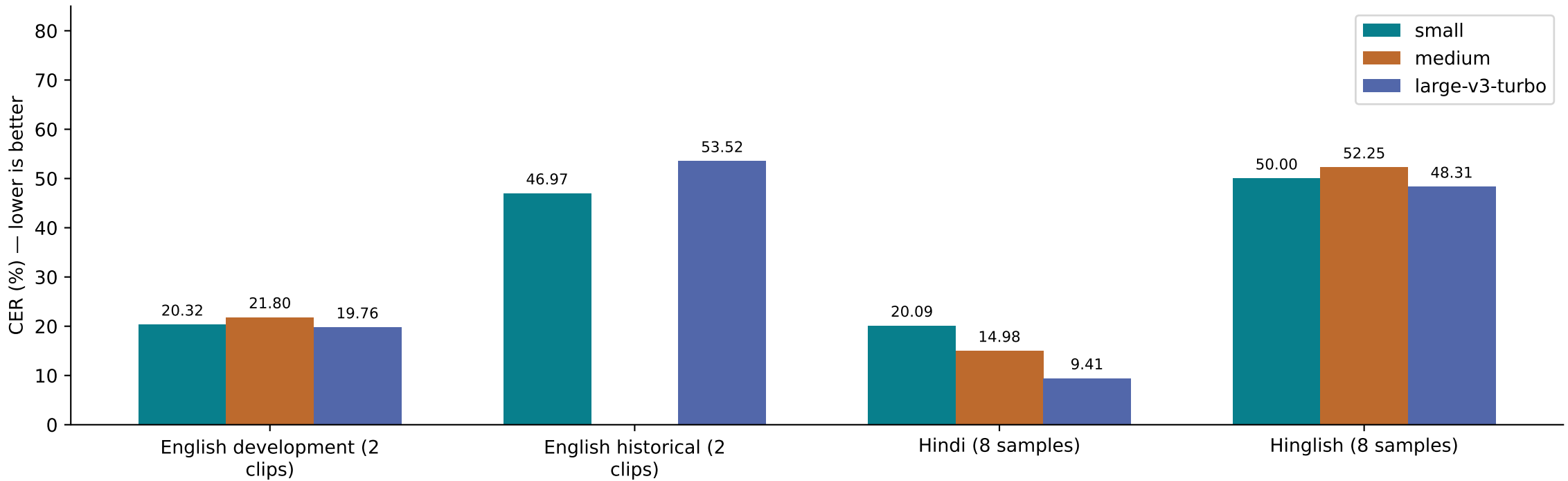
Saved discussion classifiers: local versus context features



Previously examined AMI reference-act cohort. Model-specific development selections differ; this is an exploratory comparison, not independent evidence of ensemble superiority. Installed blend is a separately selected configuration.

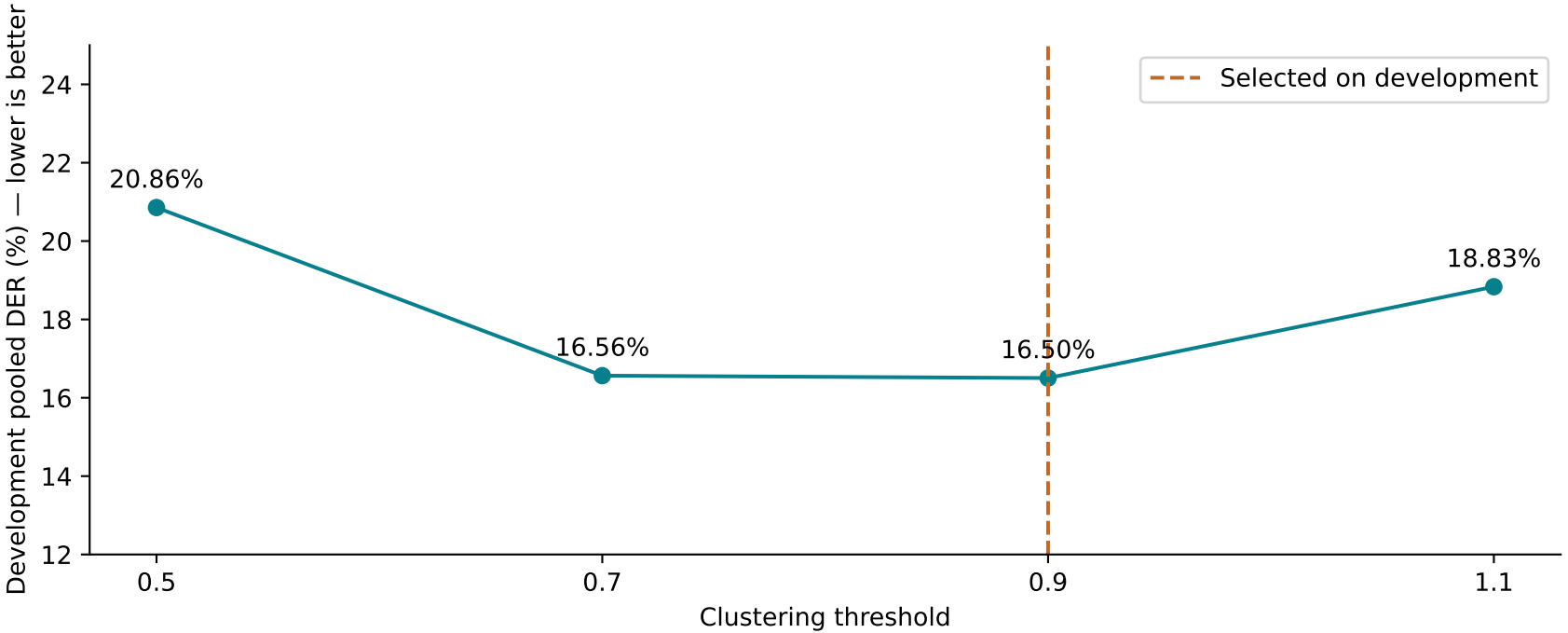


English development/historical each contain two clips; Hindi and Hinglish each contain eight samples. Missing medium historical bar means unrecorded, not zero. Cohorts are separate; no full-meeting multilingual accuracy claim.



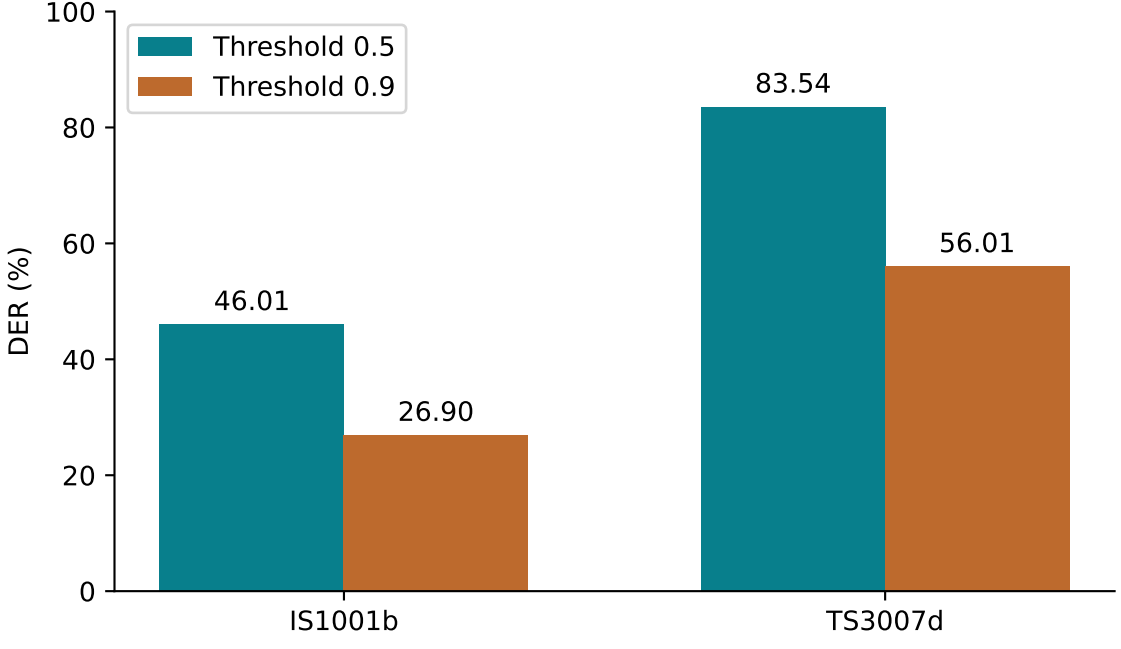
English development/historical each contain two clips; Hindi and Hinglish each contain eight samples. Missing medium historical bar means unrecorded, not zero. Cohorts are separate; no full-meeting multilingual accuracy claim.

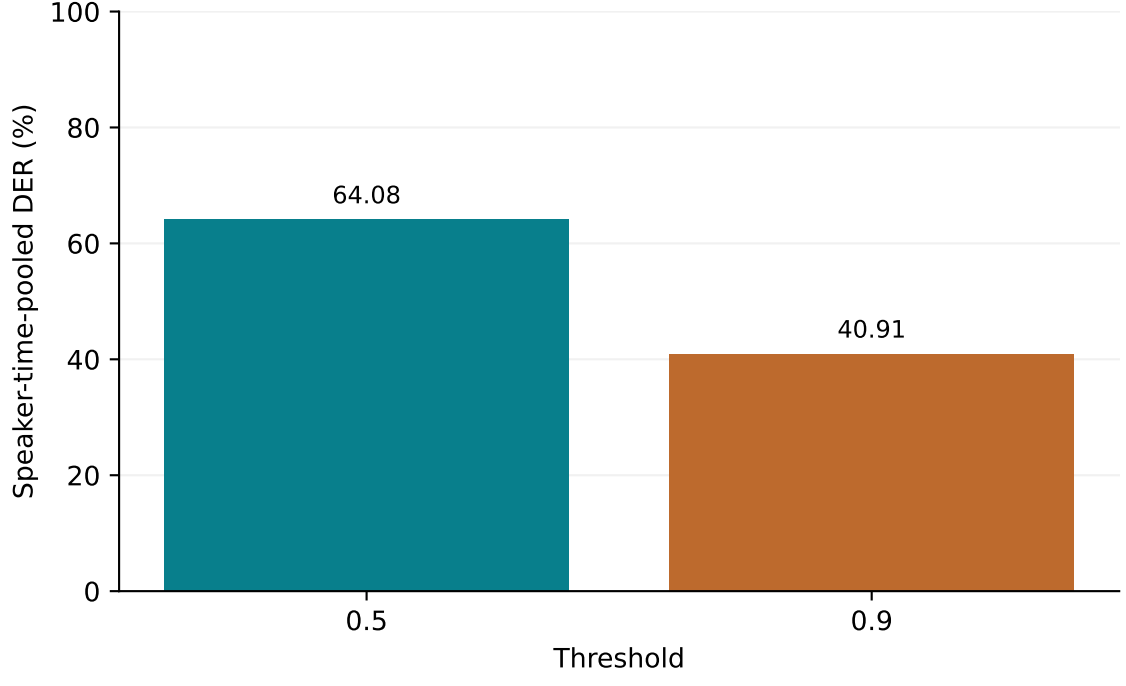
Speaker clustering threshold selection



Small development pilot only; threshold 0.9 was selected before the separate held-out excerpt comparison. This is not a full-corpus calibration curve.

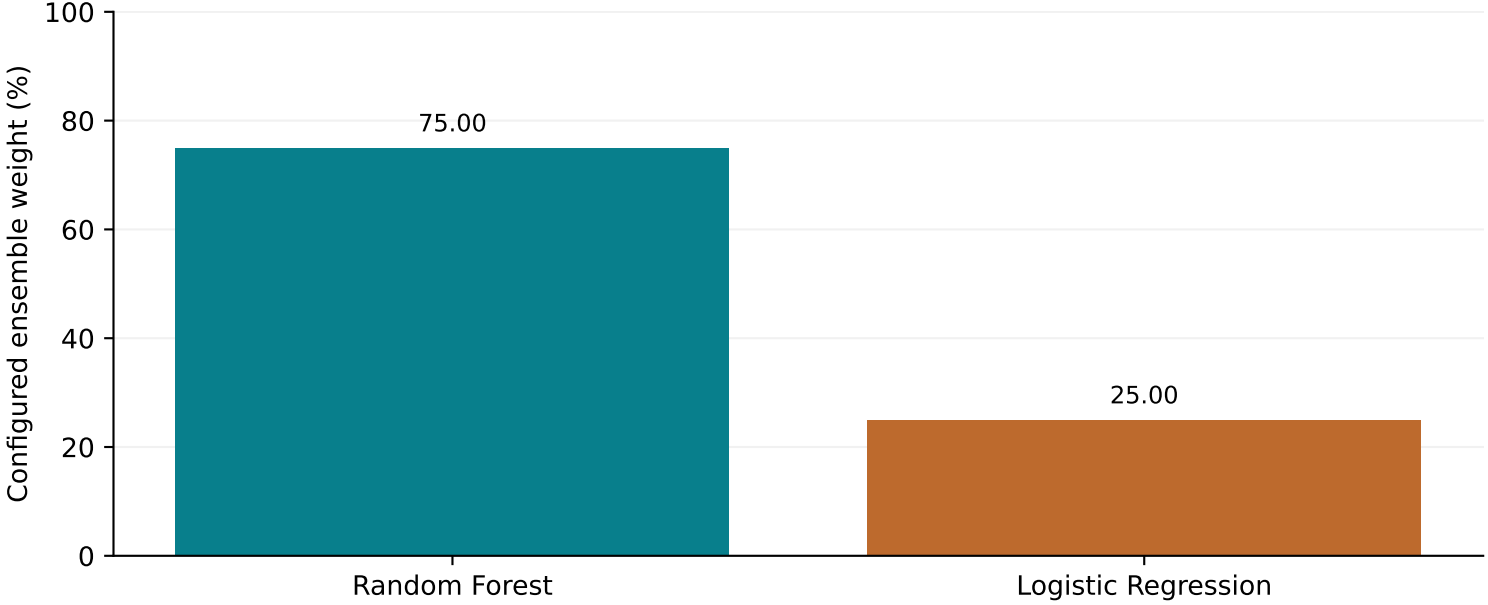
Diarization: two held-out excerpts and pooled errors





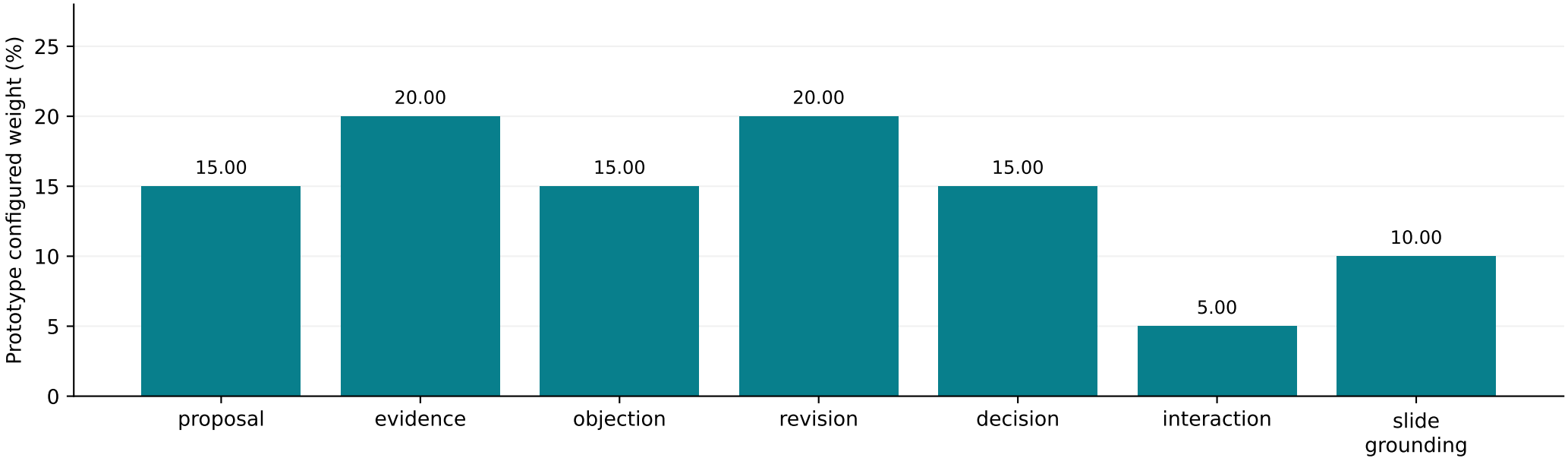
Two 120-second AMI excerpts; 0.25s collar, overlap included, optimal speaker mapping. Pool by reference speaker-time. Excerpt pilot only; no streaming or physical-device performance claim.

Discussion classifier: 75% forest + 25% logistic



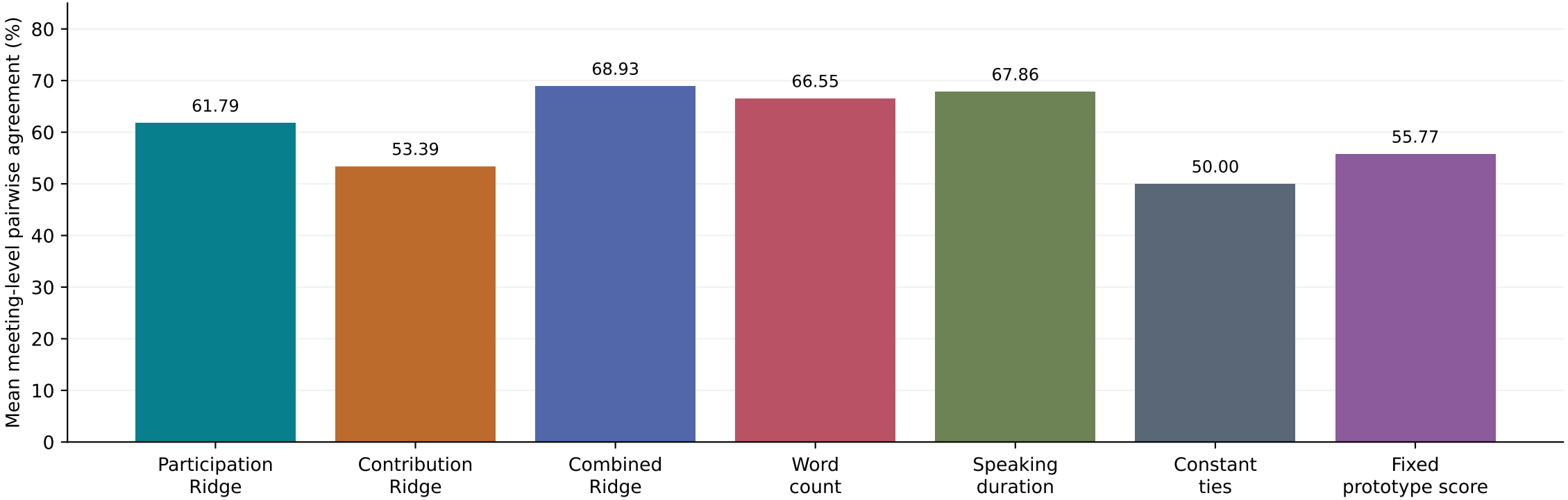
Installed model configuration, not accuracy or modality importance. Context radius 40 units; smoothing radius 5; threshold 0.35774 text-classifier outputs.

Default contribution-score weights



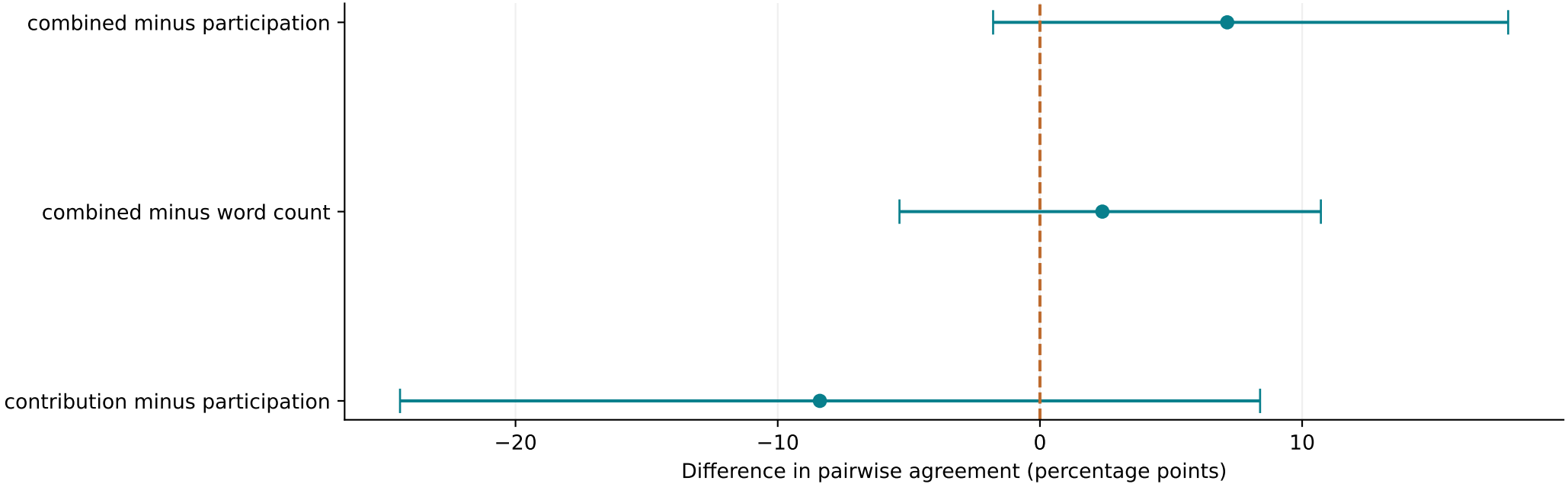
Configuration values, not empirically learned feature importance. Features are max-normalized across participants, then weighted. Missing categories stay zero; the highest participant score need not be one.

GAP research ranking: outcome alignment, not causal influence



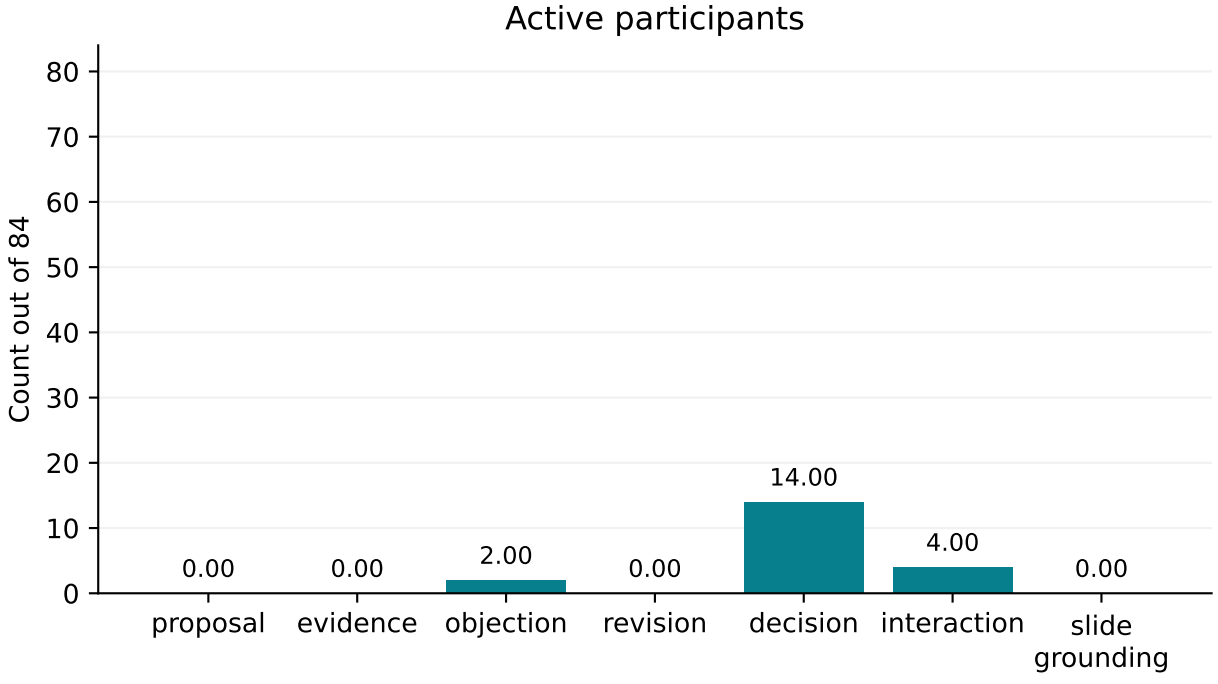
28 previously examined meetings / 84 people / 87 comparable pairs. Reference transcripts/speakers, no slides. Ridge research models are separate from the default UI score. Cross-validated exploratory point estimates.

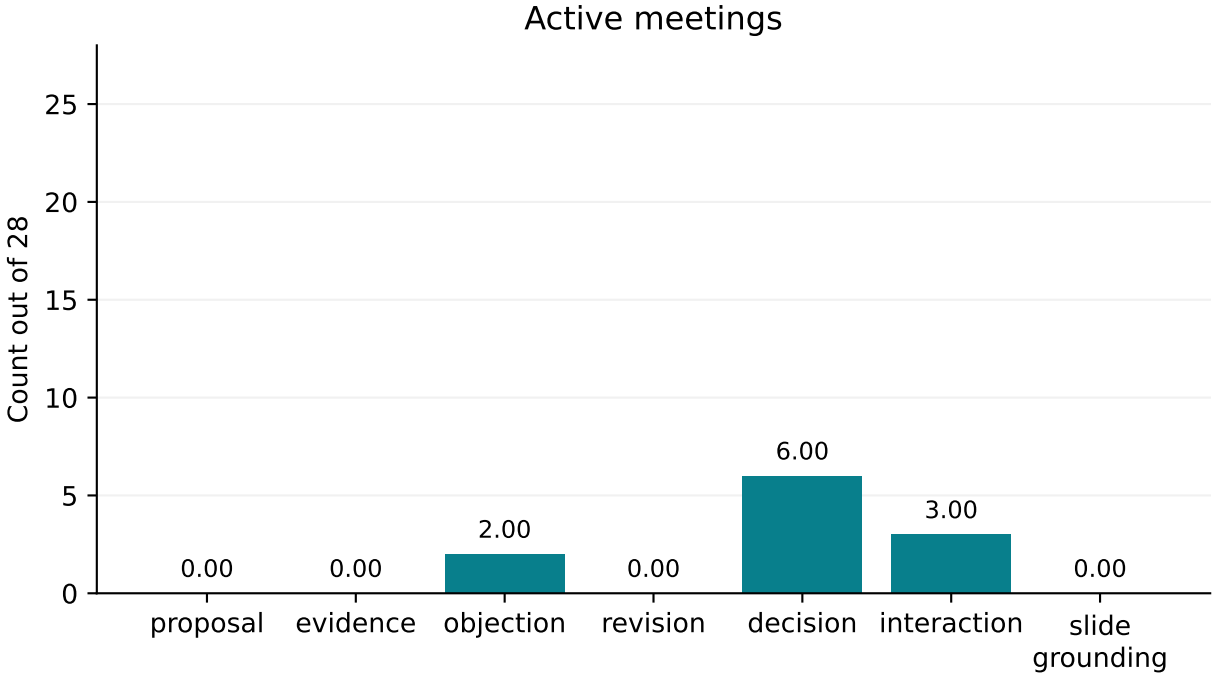
Ranking differences: descriptive intervals include zero



Intervals are copied from the saved paired bootstrap, not newly estimated. Previously examined corpus and overlapping CV training sets limit interpretation; no reliable-superiority or fresh-confirmation claim.

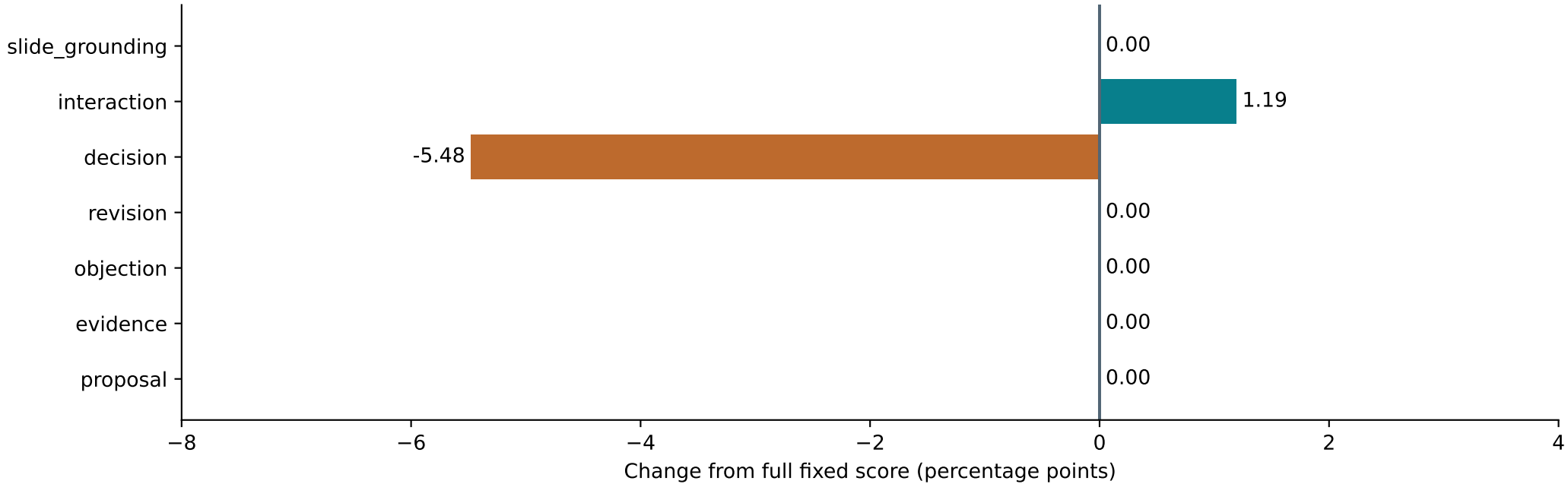
Historical extraction coverage is sparse





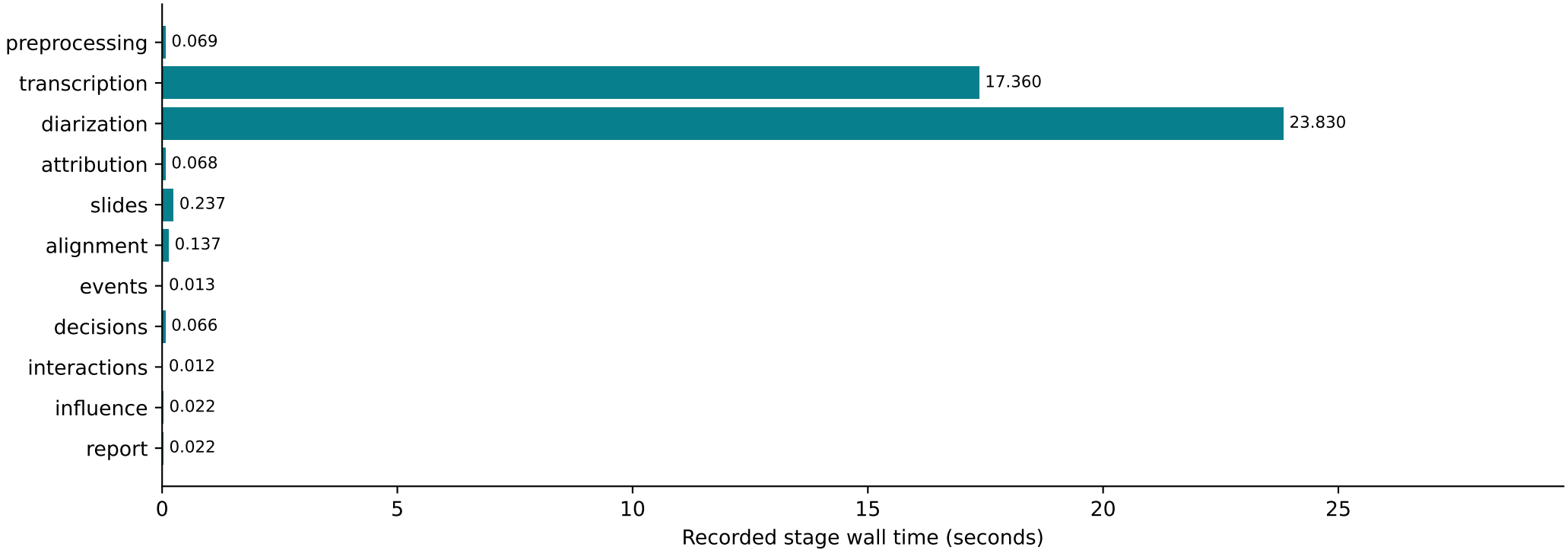
Historical snapshot: 70 / 84 participants and 22 / 28 meetings had no contribution features. No slides were supplied. Later rule fixes were not rerun as a new GAP benchmark.

Fixed-score deletion diagnostic



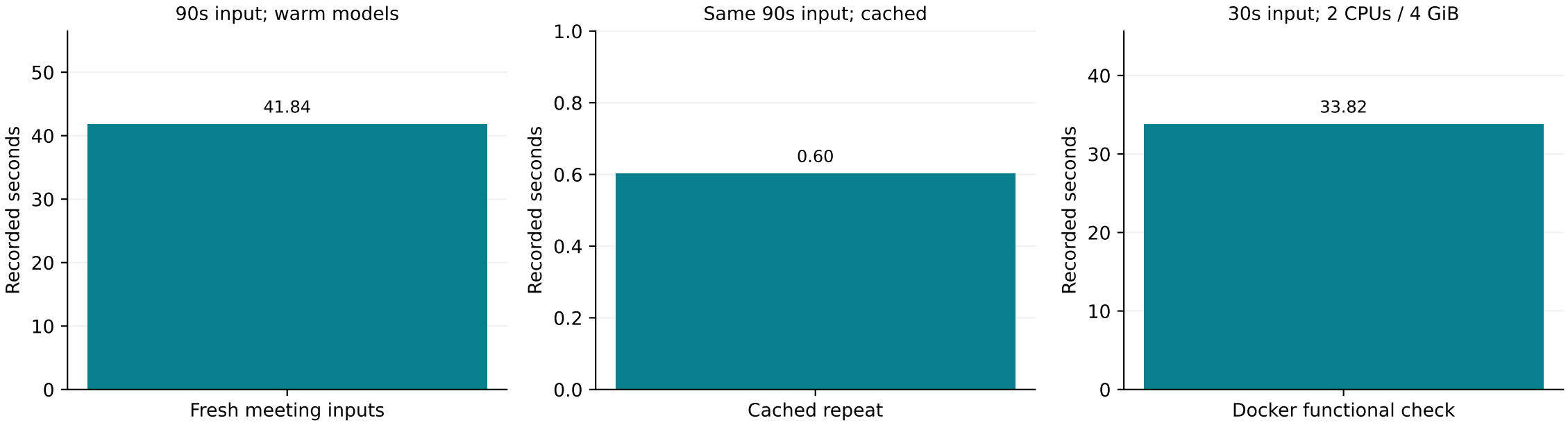
Saved delete-one-feature results on the same historical GAP representation; not retraining or causal feature importance. Inactive features can be removed without changing predictions. No confidence intervals are available.

Stage timing: 90-second English functional recording



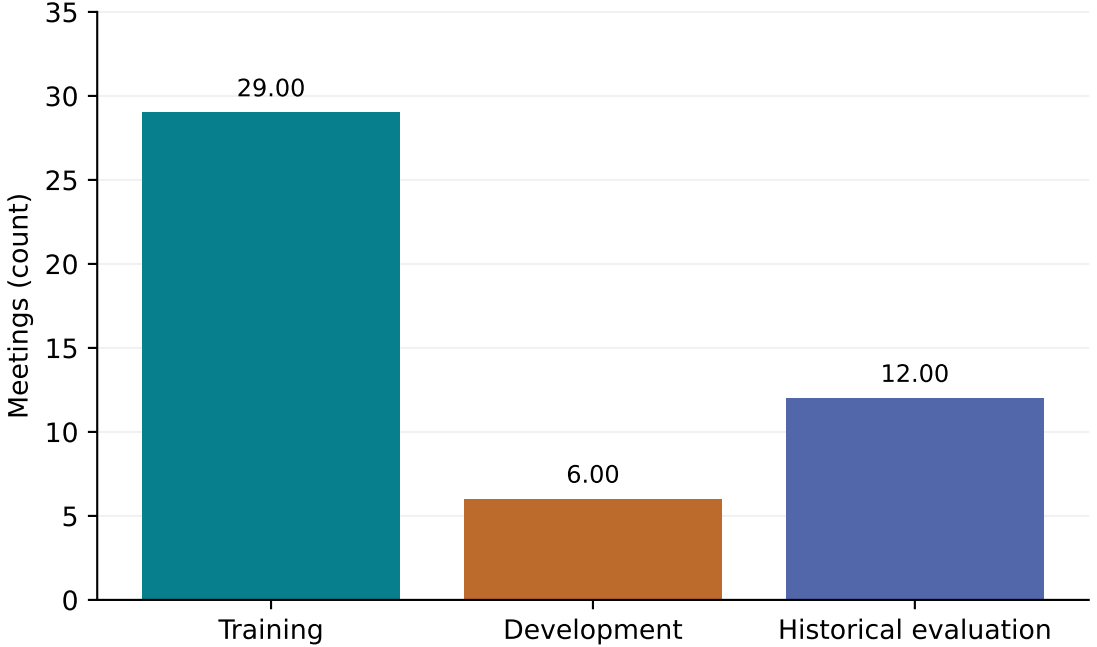
Warm models with uncached meeting inputs; two audio parts, two decks, four-speaker constraint. Excludes upload, queue, decode and final persistence. One laptop run; not a general latency benchmark.

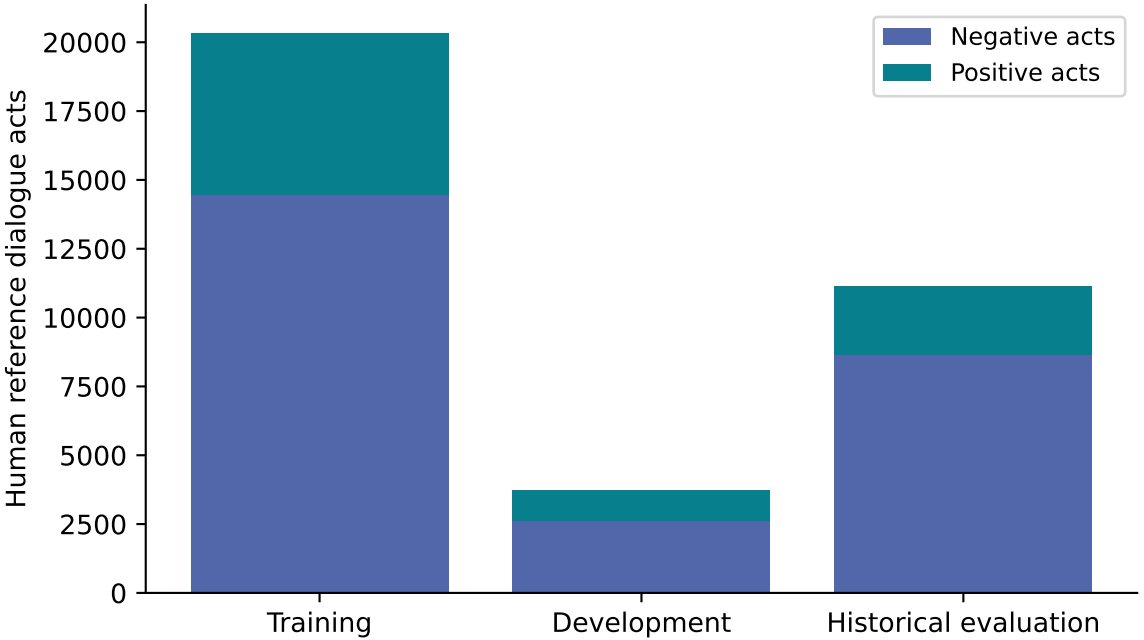
Different runtime conditions must remain separate



Independent panels use different scales and input/cache/resource conditions. These are functional timing records, not a matched comparison of models or cloud hosts. Cached repeat is not fresh inference speed.

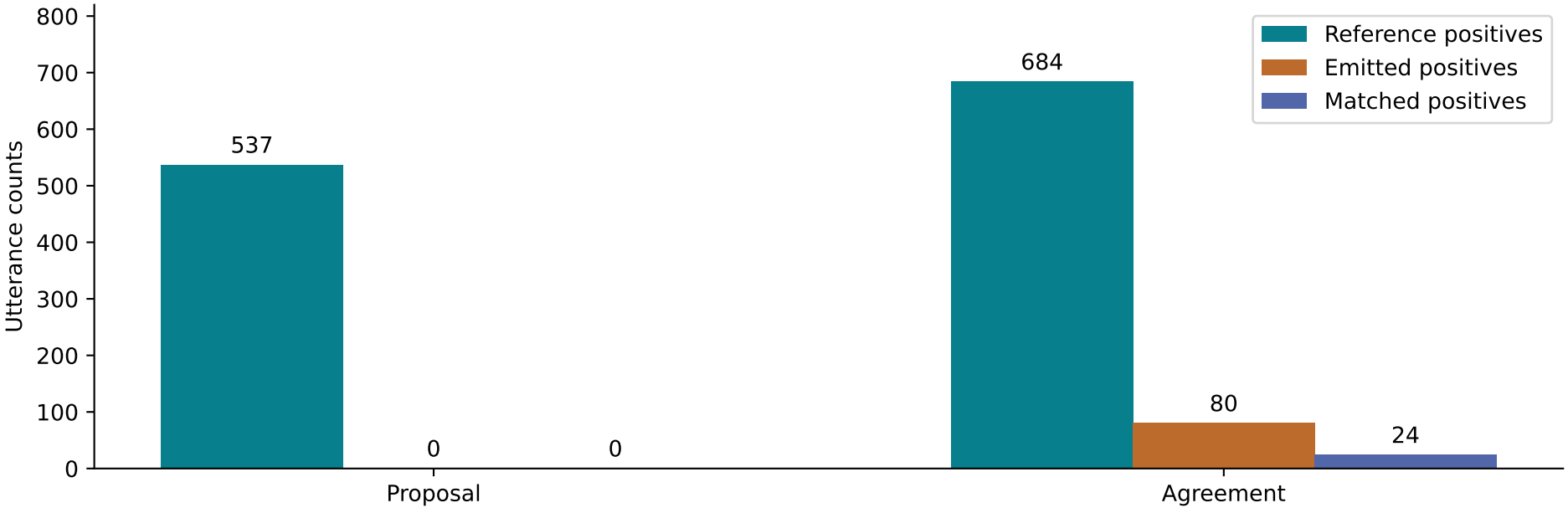
AMI discussion subset: meeting split and reference acts





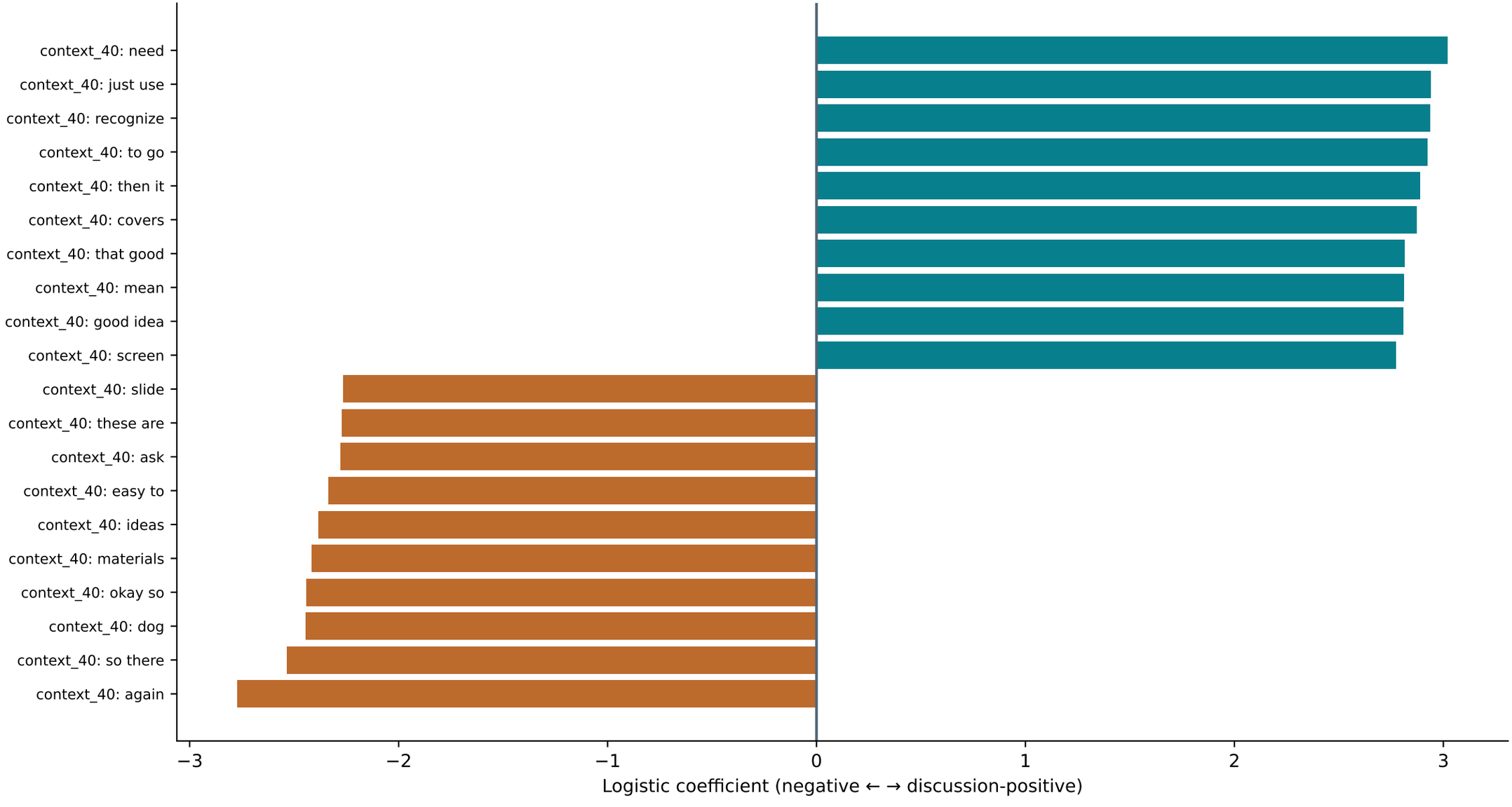
47 selected meetings, not the entire AMI corpus. Related series stay together. Historical evaluation contains 12 meetings / three series and was repeatedly examined; act counts are not independent-meeting counts.

Historical event-role coverage: a known failure finding



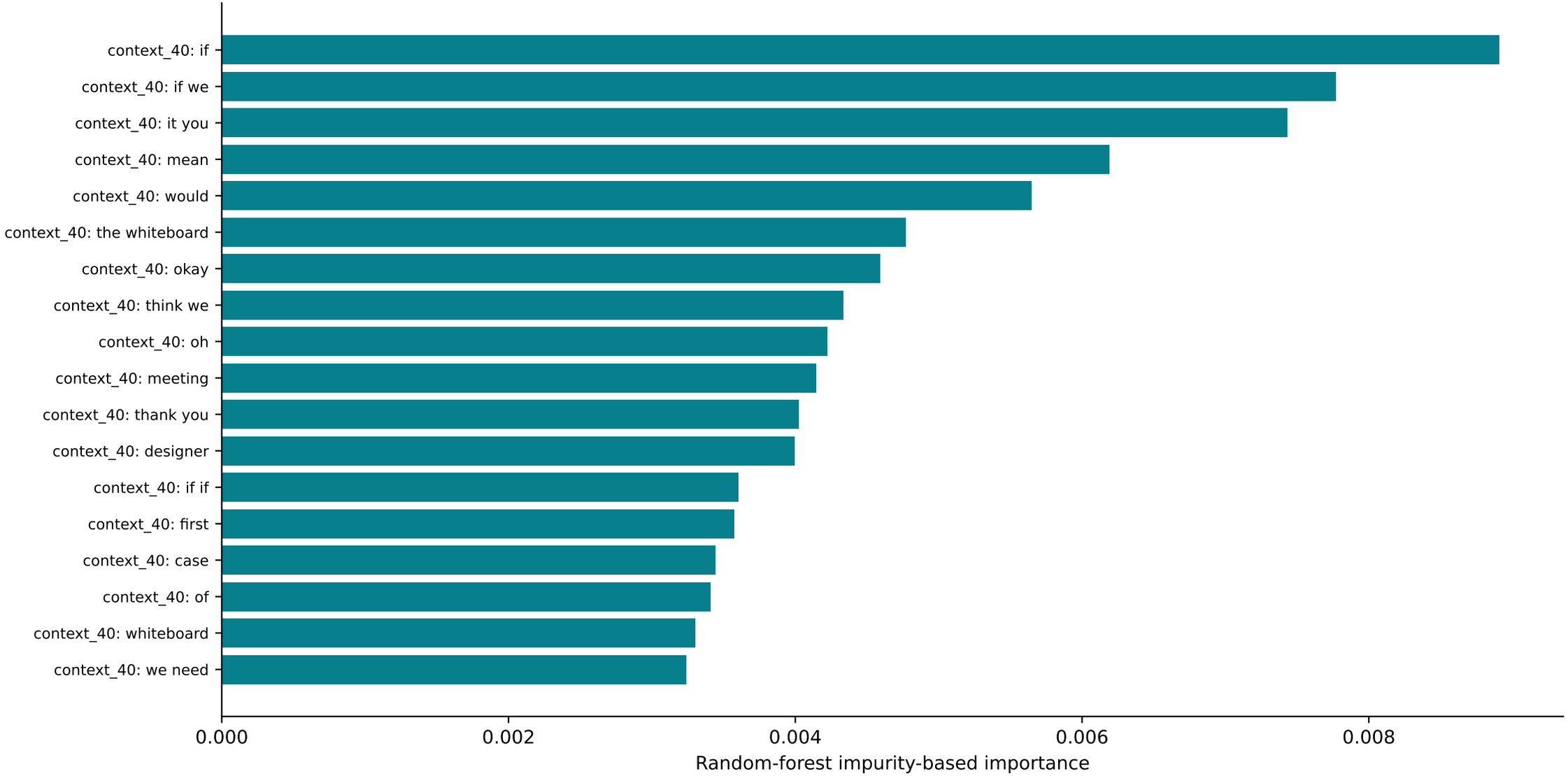
8,009 reference utterances. The evaluated historical snapshot emitted no proposals among 537 reference-positive proposal acts. Later keyword fixes do not retroactively update this diagnostic. Roles are not contribution/outcome links.

Installed logistic component: strongest signed coefficients



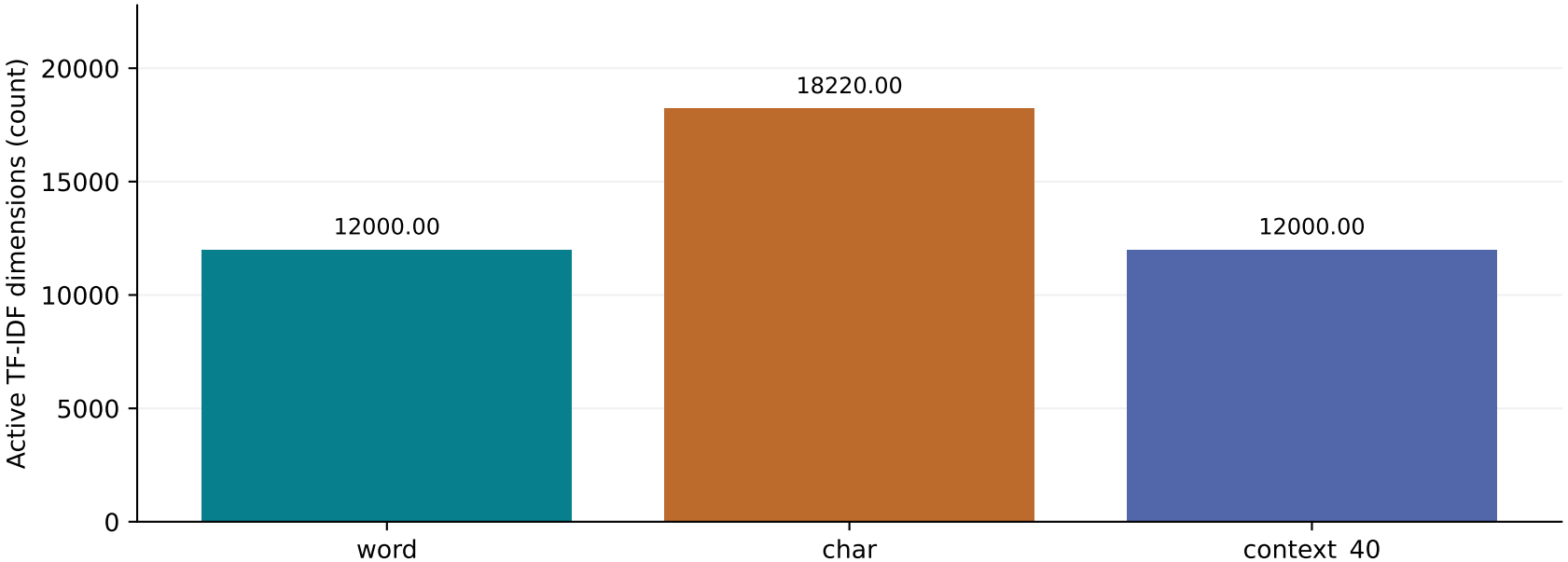
Top ten negative and positive coefficients of the actual trusted installed model. Prefix identifies word/character/context feature space.
Coefficients depend on correlated TF-IDF features; not causal explanations or full-ensemble probabilities.

Installed random forest: top impurity-based feature importances



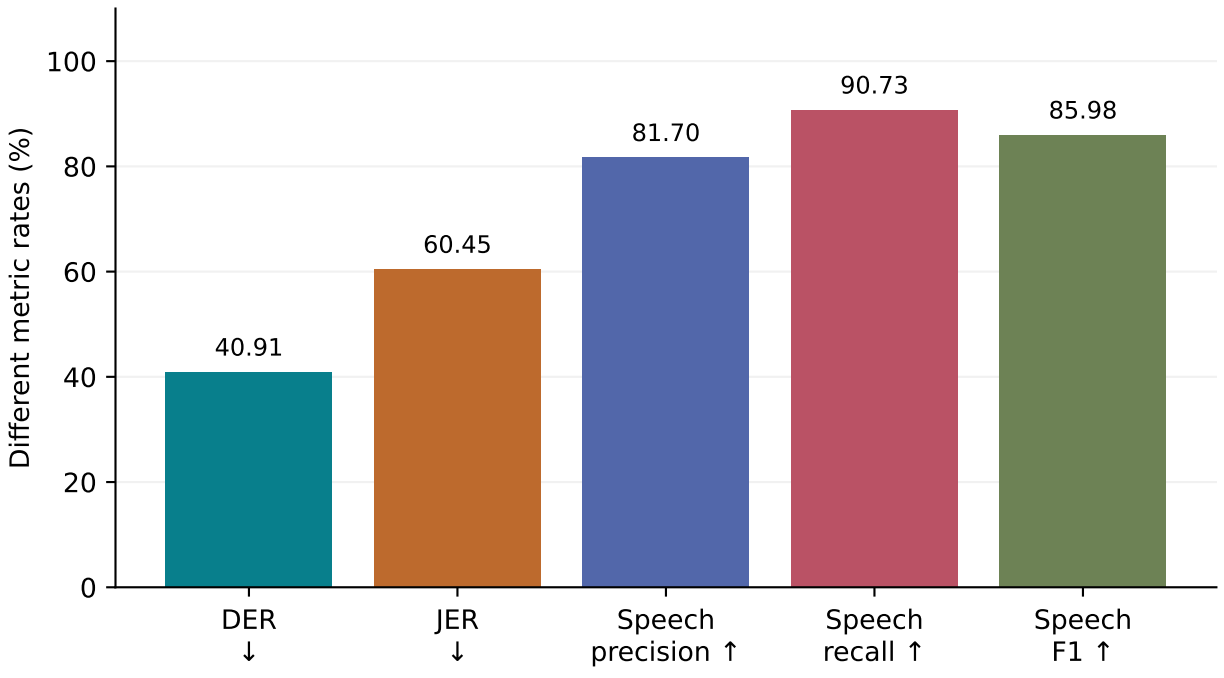
Actual forest component, not permutation importance or causal contribution. Correlated features and impurity-based biases limit interpretation; these importances do not measure the importance of physical input modalities.

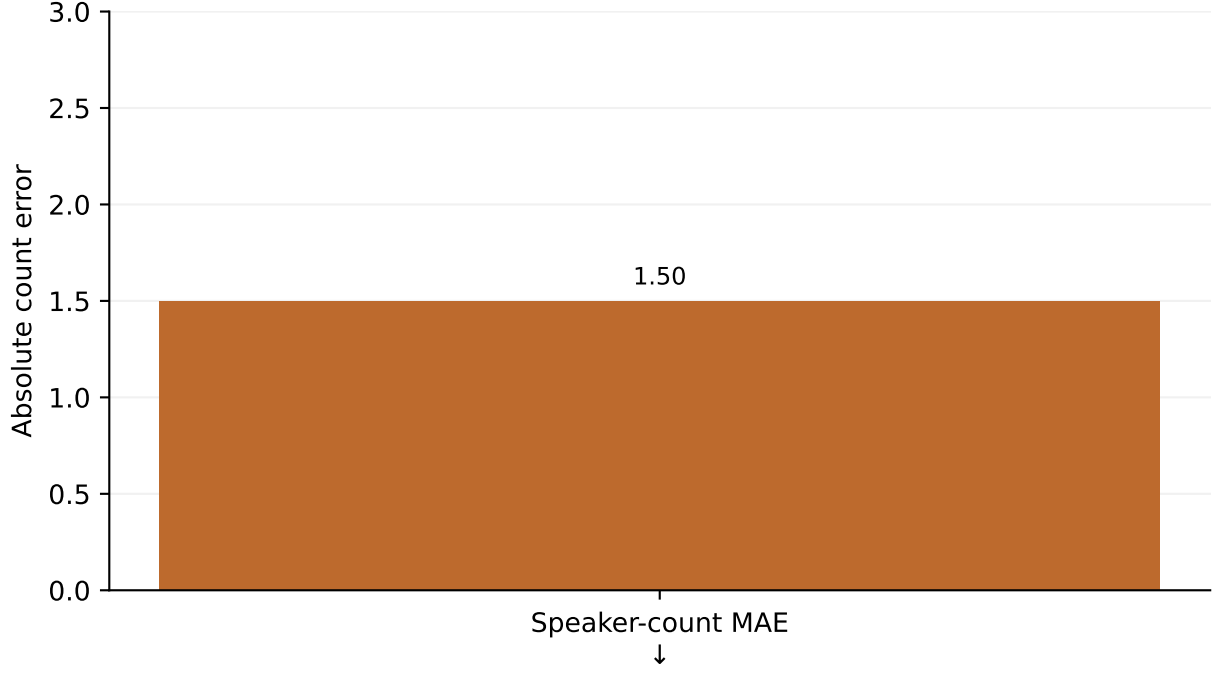
Discussion input: concatenated word, character and context features



Actual feature dimensions supplied to the installed classifiers. Stored unused context variants are excluded. Feature count is not model accuracy or an extra physical modality.

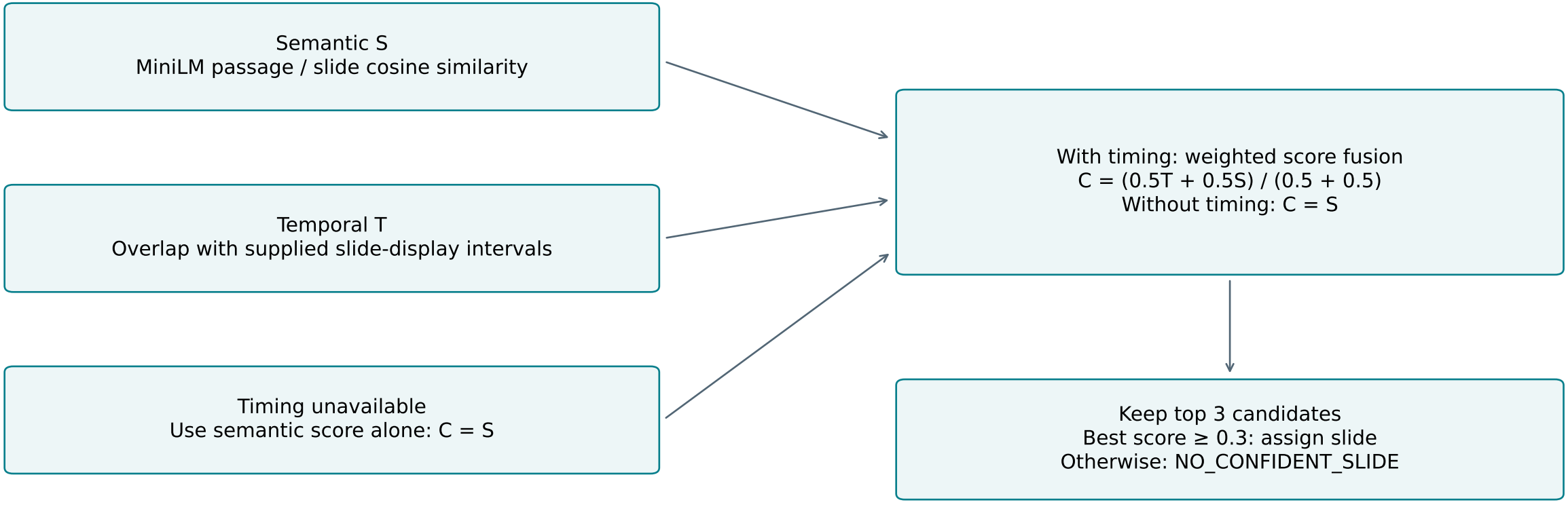
Speaker pilot: complementary diagnostic measures





Selected threshold 0.9; only two held-out excerpts. DER/JER and count MAE are lower-is-better; precision/recall/F1 are higher-is-better. Speech detection metrics concern speech activity, not word transcription correctness.

The actual alignment fusion rule and missing-timing fallback



Configuration/implementation diagram, not measured accuracy. Pure temporal-only and semantic-only settings serve as single-cue baselines. Fixed weighted cue fusion is not learned cross-attention or a neural early-plus-late hybrid architecture.